

mode revision note

October 2, 2026

finite sink for GB

In the previous model, the V/I concentration was set to be thermal-equilibrium concentration at the GB, which is a Dirichlet BC indicating the GB is a perfect sink for V/I:

$$C_V^{\text{GB}} = C_V^{\text{eq}}, \quad (1)$$

$$C_I^{\text{GB}} = C_I^{\text{eq}}, \quad (2)$$

With this Dirichlet BC, the results are mesh-dependent. With the refinement of the mesh, the defects profiles show a steeper gradient near the GB. Both the gradient and the magnitude of the solute segregation are higher than coarse mesh. These results are also observed in other RIS models using Onsager equation. **[RN486]**